

Information loss of fixed linear health indicators as a control problem: simplex geometry and reachability limits

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Running title: Health indicators as a control problem

Abstract

Health indicators combine monitoring channels with fixed weights, yet the combination most informative about remaining life rotates as degradation mechanisms progress at different rates, so the indicator discards information. We treat that loss as a control problem: the operating point enters the degradation dynamics and steers the informative direction. For a linear indicator, positive degradation rates and Gaussian remaining life over a time-shift population, the loss depends on the indicator only through one misalignment angle, a distance on a statistical simplex where degradation is an input-modulated replicator flow. Zero loss confines the state to a segment fixed by the indicator and is attainable exactly when admissible inputs align the indicator along it; with fewer independent inputs than mechanisms less one, almost no indicator attains it. The loss depends on the record's Fisher information only jointly with the indicator's own. The best achievable loss solves an optimal control problem with a ratio objective and a controlled terminal time, the failure time. On one synthetic plant, a myopic steering policy lowers the information loss objective 47-fold against the best constant operating point with its own indicator, within numerical error of the optimum. Across 299 random plants of the separable model, the reachable-set width, known before design, provides a moderate screening signal for when steering pays (area under the receiver-operating curve 0.73). Public data show the rotation in measured battery cells, leave actuation untested, and reject the separable form jointly with its sensor pairing on simulated turbofans.

Keywords: fractional optimal control; reachable sets; information geometry; health indicator; prognostics; re-

maining useful life

1 Introduction

Condition monitoring rarely feeds raw measurements to a life-prediction algorithm. A vibration channel, a temperature, an impedance and a capacity reading first go into a single scalar, the health indicator, and the life-prediction algorithm sees only that scalar [1, 2]. The compression is almost always fixed: the designer picks one set of weights at design time, and every unit carries those weights for its whole life, because firmware embeds them, certification freezes them, or re-tuning them in the field is impractical.

Fixing the compression has a cost. Real components degrade through several mechanisms at once, and those mechanisms do not proceed in step: an early-life process may dominate for the first third of the record and then saturate, while a late-life process accelerates. The combination of channels that carries the most information about remaining life therefore rotates as the component ages. A fixed indicator is aligned with that combination at best at one moment, and is misaligned for the rest of the record; the information it discards is not recoverable downstream, however good the prediction algorithm that follows [3].

The established response is to make the map better: choose a different criterion for it, let it switch between segments of life, or learn it from more run-to-failure data [2, 4]. All of these take the degradation as given and work on the compression. This paper acts on the degradation itself, through the operating point.

The rotation is not a property of the instrumentation. The degradation physics produces it, and that physics responds to stress: temperature, load and duty cycle change the relative rates of the competing mechanisms, not only the overall speed. Running a cell hotter does not only age it faster; it also shifts the balance between the mechanisms, and therefore moves the direction in which information about remaining life arrives.

The operating point is, in most applications, something the operator chooses. So the misalignment be-

tween a fixed indicator and the informative direction is not fixed by nature. It is a function of the input, and the operator can steer it.

That reframing turns a data-analysis question into a control question, and control questions come with sharper answers than design heuristics do. One can ask what misalignments are reachable at a given operating envelope, whether zero loss is attainable at all, how many independent inputs that would take, and what the best achievable loss is when it is not. These are the questions this paper answers.

In the terms of nonlinear control, the plant is a positive system whose input enters through the stress factors rather than additively, and the indicator is a fixed static functional of its measurements. Zero loss asks that the $d - 1$ components of the information-carrying vector orthogonal to the indicator vanish along the whole trajectory: an output-zeroing problem, with direct feedthrough, for $d - 1$ outputs and m inputs [5]. Holding those outputs at zero pins the direction of motion, so the state is confined to one segment fixed by the indicator, and zero loss is attainable exactly when an admissible input aligns the indicator at every point of it; the count of independent inputs is necessary for this and not sufficient. The best achievable loss is an optimal control problem whose objective is a ratio and whose horizon, the failure time, is set by the input, so a fixed-horizon formulation misrepresents it, and whether a policy that aligns the indicator as closely as possible at each instant is optimal has to be computed rather than assumed. Section 2.5 states the three problems formally.

They sit against five bodies of work. Four address parts of the problem and the fifth supplies the geometry used in Section 3; to our knowledge none of them addresses the position taken here.

The prognostics literature treats the compression as a design problem to be solved from data: searching genetically over candidate features against monotonicity and trendability criteria [2], fusing the sensors linearly into a composite health index whose weights are fitted once to a degradation model [6] and analysing that fusion statistically [7], learning it with a recurrent network [8], or surveying the resulting pipelines [1, 9]. The composite indices of [6, 7] are fixed linear maps of exactly the kind studied here. Two limits on that programme are known, information-theoretic bounds on what any fixed compression can retain [3] and the number of run-to-failure units needed to learn one reliably [4]. The first is our own, and the boundary is worth drawing explicitly: it takes one degradation trajectory as given and asks how much information a fixed compression must lose along it. The question here is the complementary one, whether that trajectory can itself be steered so that the same fixed map loses less. Both prior limits hold the trajectory fixed; neither asks whether it can be changed.

The predictors downstream are overwhelmingly stochastic — Wiener, gamma and inverse-Gaussian degradation processes with first-passage failure [10, 11],

reviewed for remaining-life estimation in [12, 13] and embedded in system-level design methodology in [14]. That literature models how damage evolves, not how the measurement should be compressed before the model is applied, which is the question upstream of it.

A separate literature does treat the operating point as a decision variable, but to extend remaining life rather than to preserve estimability [15, 16]; the two aims can conflict, and Section 6 exhibits an instance in which they disagree. Choosing inputs to make a state easier to estimate is itself classical, going back to dual control [17], and its modern forms shape a scalarisation of the Fisher information matrix [18–20]. Our Proposition 4.7 shows that the quantity at issue here is not a function of the record’s Fisher information, which such designs score, but of its ratio to the indicator’s own, so those methods apply once they score the indicator.

The fifth body is information geometry. The reformulation of Section 3 rests on the Fisher–Rao geometry of the probability simplex and on the Bhattacharyya coefficient [21, 22], and the flow it produces is the replicator equation of evolutionary dynamics [23, 24]. These are standard objects; what is new here is their identification with the informative direction of a degrading component, and the fact that the fitness driving the flow is actuated.

The contributions are as follows.

1. The information a fixed indicator discards is an exact function of one misalignment angle (Lemma 2.2). Placing the channels’ shares of the Fisher information on a statistical simplex turns the degradation into a replicator flow whose rates the operating point modulates, and the optimal indicator into a barycentre of the degradation path (Section 3).
2. Zero loss confines the state to one segment fixed by the indicator, and is attainable exactly when every state on that segment admits an admissible aligning input (Section 4). With fewer independent operating variables than mechanisms less one, almost every indicator fails. On a test plant, supplying the missing input collapses the peak reachability floor from 4.52° to zero, with the aligning input inside the envelope along the whole segment, so the rule prices instrumentation rather than only bounding it. The count of inputs is necessary for generic data and not sufficient: the envelope condition along the segment decides. Taking logarithms makes every reachable set a translate of one fixed set, which turns that condition into a containment; the exceptions to the count exist for any number of mechanisms and are too fragile to build on (Section 4.1). The count separates two objectives that are easily conflated: holding the misalignment at a constant level is a single scalar condition and generically needs one input, whereas freezing the informative direction outright is $d - 1$ conditions. That gap is

Table 1: Where this paper sits. The columns are the axes that separate these literatures, not a ranking: each body of work answers a question the others do not ask, and the last column is a restriction this paper imposes on itself rather than an advantage over the others. “Input alters degradation” means the operating point enters the degradation dynamics; “direction” means the informative direction of Section 2 is itself an objective.

	input alters degradation	lifetime is an objective	direction is an objective	indicator fixed, scalar
Indicator design [2, 6–8]	—	—	—	✓
Fixed-indicator limits [3, 4]	—	—	—	✓
Stochastic life models [10, 12]	—	—	—	—
Health-aware control [15, 16]	✓	✓	—	—
Observability-aware [17, 18, 20]	✓	—	partly	—
This paper	✓	secondary	✓	✓

the difference between useful steering and lossless steering, and an installation with one controllable variable can have the first without the second.

3. The loss is not a function of the Fisher information the record carries about remaining life: at any fixed value of it the loss can take any value up to its maximum, so input design that scores the record cannot rank inputs by the loss. The loss is fixed by the record’s information together with the indicator’s own, which is what such design should score instead.

4. The best achievable loss is the value of an optimal control problem whose objective is a ratio, and whose terminal time is itself controlled (Section 5); we bracket its value numerically, find it indistinguishable on the test plant from a myopic policy that ignores the controlled horizon, and do not certify it. The actuation authority, measured by the width of the reachable set and computable before any indicator is designed, provides a moderate screening signal for the regime in which steering can have substantial value; on 299 randomly drawn plants it ranks them as well as its ratio to the drift of the informative direction and fits them better, and the drift matters only in combination with it (Section 6.7).

5. Robustness to model error is measured rather than assumed (Section 6.9): an indicator designed from activation energies in error by 10% costs a median factor 3.10 in loss, while an error of 2% costs almost nothing, and the plants that escape the counting rule are knife edges on which one percent of shape error makes steering worse than not steering.

6. The premises are separately testable on public run-to-failure data, and Section 7 tests three in turn whether the informative direction rotates, whether the operating point moves it by acting on the damage, and whether the separable stress model, read through the pairing of channels with mechanisms, survives. The first holds on measured fleets. The second is untested, because no public fleet lets the operating point act on the damage; the simulated turbfans show the observed direction moving with

the regime through the sensor response alone. The third is rejected, jointly. That section therefore reports which hypotheses withstand contact with data rather than demonstrating that the method works.

The results do not all rest on the same assumptions, and it is worth separating them before any of them is stated. Lemma 2.2, Proposition 2.3 and Theorem 4.1 need only the observation model of Section 2: a fixed linear indicator on whitened channels, and a scalar latent. Lemma 4.4 and Proposition 4.5 need more, namely that the input enters each mechanism through a single separable factor, and Section 7 reports that the highest-fidelity turbofan data reject that form jointly with the pairing of channels with mechanisms in (5), on which the positivity of the informative direction also rests. The empirical readings of Section 7 need more again, since estimating one informative direction from a fleet presumes the units differ by a time shift rather than by their degradation coefficients. Fig. 6 is the map, and the reader who wants to know which branch a given result sits on should consult it rather than infer it.

Section 2 states the model. Sections 3 to 5 develop the theory, Section 6 exercises the three of its claims that are stated sharply enough to be refuted, and Section 7 reports what public fleet data can and cannot establish about the premises the theory rests on.

Notation S^{d-1} is the unit sphere in $\mathbb{R}^d$, $\|\cdot\|$ the Euclidean norm and $\angle(a, b) \in [0, \pi]$ the angle between non-zero vectors a and b . The state x lies in the closed positive orthant and the input u in a compact set $\mathcal{U} \subset \mathbb{R}^m$; an admissible input is a measurable function $u(\cdot)$ with values in $\mathcal{U}$ for which (2) has an absolutely continuous solution that reaches the condemnation limit in finite time, so the failure time T_f is a functional of the input. The indicator is a unit weight vector $\mathbf{v}$; from Section 3 on it is identified with its share vector q , $q_i = \mathbf{v}_i^2$, through the isometry between the simplex and the positive part of S^{d-1} , and q denotes both. The misalignment $\angle(\hat{\mathbf{d}}, \mathbf{v})$ is written ψ in Section 2 and β thereafter.

2 Problem formulation

This section states the degradation model, says precisely what a prognostic algorithm is being asked to estimate and where the uncertainty in that quantity comes from, derives the direction along which information about it arrives, and defines the fixed indicator whose information loss the rest of the paper studies. Readers familiar with [3] will recognise the model; the derivation is given in full here because the geometry of Section 3 depends on several of its details.

2.1 Degradation driven by a controlled operating point

A component degrades through d distinct mechanisms acting at once. In a lithium-ion cell these might be loss of cyclable lithium, growth of the solid-electrolyte interphase and mechanical fatigue of the electrode; in a rolling bearing, spalling, wear and lubricant degradation. We write $x_i(t) \geq 0$ for the damage accumulated by mechanism i up to time t , and collect these into the state $x = (x_1, \dots, x_d)$.

One externally chosen operating point drives all mechanisms. It collects the temperature, the mechanical or electrical load, and the duty cycle at which the operator runs the component. We write that operating point as

$$u(t) \in \mathcal{U} \subset \mathbb{R}^m, \quad (1)$$

with $\mathcal{U}$ compact, representing the operating envelope the application permits. Throughout this paper u is the control input: it is the decision variable, and the question we ask is what choosing it well can buy. The number of independent inputs, m , turns out to decide what is achievable, and it is usually much smaller than the number of mechanisms d .

The damage states evolve according to

$$\dot{x}_i = \varphi_i(u) h_i(x_i), \quad i = 1, \dots, d, \quad (2)$$

in which the two factors play different roles:

- $\varphi_i(u) > 0$ is the stress-acceleration factor of mechanism i : how much faster that mechanism runs at operating point u than at a reference point. It carries all the dependence on the input.
- $h_i(x_i) > 0$ is the intrinsic shape of mechanism i : how its rate depends on the damage it has already done. It carries all the dependence on the state.

This separation is the standard accelerated-life form and covers the stress models in routine use [25]. In the Arrhenius model the input is absolute temperature and $\varphi_i(u) = A_i \exp(-E_i/(ku))$, with $A_i > 0$ a pre-exponential constant, $E_i > 0$ the activation energy of mechanism i and k Boltzmann's constant. In the inverse-power-law model the input is a mechanical or electrical stress and $\varphi_i(u) = \kappa_i u^{n_i}$, with $\kappa_i > 0$ a scale factor and $n_i > 0$ the stress exponent. Choosing

h_i recovers the exponential, power-law and stretched-exponential degradation classes [3]; the results below use only $\varphi_i > 0$ and $h_i > 0$.

The within-unit dynamics are deterministic, where most of the degradation literature places the randomness inside a single unit's path [10, 11, 13]. We put it between units instead (Section 2.2). The object of study is the direction along which information arrives, which is fixed by the relative rates of the competing mechanisms: a within-unit diffusion would jitter that direction without altering its systematic motion, whereas dispersion in the rate parameters across a population is what makes remaining life uncertain and therefore worth estimating.

We measure damage on a scale normalised so that mechanism i exhausts the component's tolerance when x_i reaches unity. A component is withdrawn at a condemnation limit at or below that point, so we write the end of life as the first passage of any one mechanism to a threshold $x_f \in (0, 1]$:

$$T_f = \inf\{t \geq 0 : \max_i x_i(t) \geq x_f\}. \quad (3)$$

No result below depends on the value of x_f , which enters only by fixing where a trajectory stops. The computed quantities of Section 6 do depend on it, and the value used there is stated with the rest of the test plant.

2.2 What is to be estimated, and why it is uncertain

The quantity a prognostic system must report is the remaining useful life

$$R(t) := T_f - t, \quad (4)$$

the time left before failure.

We must say where the uncertainty in R lies, because (2) is a deterministic differential equation: for one component whose initial damage and parameters were known exactly, T_f and hence R would be known exactly, and there would be nothing to estimate. The uncertainty is between components, not within one. Units built to the same specification differ in initial damage and in the coefficients of (2), which is the unit-to-unit variability that reliability engineering treats as the primary source of scatter in life data [25]. A prognostic algorithm observes one unit drawn from that population and must infer its remaining life.

We therefore treat R , at any given observation instant, as a random variable over the population of units, with prior variance $\text{Var}[R] > 0$. This variance is the uncertainty the algorithm exists to reduce, and it is the reason the estimation problem is not trivial. Everything below is written for a nominal trajectory of (2), with the fleet spread entering only through $\text{Var}[R]$.

2.3 Observations and the informative direction

Each damage state reaches the record through its own noisy channel,

$$y_i = x_i + \sigma_i \epsilon_i, \quad i = 1, \dots, d, \quad (5)$$

with $\sigma_i > 0$ the noise level of channel i and ϵ_i independent standard normal variables. The record available to the algorithm is the collection of these measurements over the observation window.

Equation (5) pairs each mechanism with a channel of its own. Condition monitoring in general returns $y \equiv H(x, u) + \epsilon$, with sensors that mix mechanisms, and the pairing is an assumption rather than a description. It is load-bearing: the positivity of $\mathbf{d}$ that Remark 3.1 shows the results to need follows from $\varphi_i > 0$ and $h_i > 0$ in these coordinates and need not survive an arbitrary H . Where H is known, invertible and positivity-preserving the construction carries over to $H^{-1}y$; the general case does not reduce to it.

Two assumptions carry the sensitivity, and both are stated here rather than left to be inferred.

The prior on R We take R to be Gaussian over the population with variance $\text{Var}[R]$. Linearising the record about the nominal trajectory, following [3], then makes the record and R jointly Gaussian, and the mutual informations below are exact. Linearisation alone would not do it: with a non-Gaussian prior of the same variance the Gaussian value is an upper bound on what a channel carries, and Lemma 2.2 becomes an inequality rather than an identity. The information geometry that follows is exact under this Gaussian prior and the time-shift population described next; it is not a model of population variability in general.

How units differ We take the population spread to be a reparametrisation of a single degradation path: units differ in where they sit on the trajectory of (2), not in its shape. This is what makes R a coordinate along that path, and it is what the next display uses. Dispersion in the coefficients of (2) is a different matter, because it moves $\partial x / \partial \theta$ rather than sliding along $\partial x / \partial R$, and Remark 2.1 records what it costs.

The sensitivity of channel i to remaining life then follows from (4): because R falls at unit rate as t advances,

$$\frac{\partial x_i}{\partial R} = -\frac{\partial x_i}{\partial t} = -\varphi_i(u) h_i(x_i). \quad (6)$$

Remark 2.1 (what the time-shift assumption costs) Equation (6) is exact when the population spread slides along one path, and only then. On the plant of Section 6.1, with units instead drawn by dispersing the degradation coefficients, the direction that the population regression of x on R actually returns stands about 45° from $-\varphi(u)h(x)$, across two families of dispersed coefficient and every spread tried; a control whose units

are started on the reference trajectory, where (6) holds by construction, returns the same regression to within 0.2° . The misalignments this paper is about are a few degrees. Extending the construction to populations dispersed in θ , where the observation carries scatter that no value of R explains, is open.

Dividing each channel by its own noise level puts them on a common scale, and taking magnitudes gives the whitened signal direction

$$\mathbf{d}(t) = [\sigma_i^{-1} \varphi_i(u) h_i(x_i)]_{i=1}^d, \quad \hat{\mathbf{d}} = \mathbf{d} / \|\mathbf{d}\|. \quad (7)$$

In words: $\mathbf{d}$ says how strongly, and in what combination of channels, the measurement responds to a change in remaining life. Its direction $\hat{\mathbf{d}}$ is the *informative direction*, the combination a health indicator has to be aligned with and the object named informally in Section 1. Its length $\|\mathbf{d}\|$ is the strength of that signal. The two objects carry the paper: $\|\mathbf{d}\|$ is how much information arrives and $\hat{\mathbf{d}}$ is where it arrives from. The loss of a fixed indicator depends on the second through the misalignment, and on the first only through a weight that saturates as the signal grows (Lemma 2.2). Experiment design in the observability tradition scores the first, and Proposition 4.7 shows that the first does not determine the loss.

We use two properties of (7) repeatedly below.

Positivity Every entry of $\mathbf{d}$ is strictly positive, because $\varphi_i > 0$ and $h_i > 0$ in (2). This is a consequence of the degradation model rather than an assumption imposed for convenience, and Remark 3.1 shows that the results below would fail without it.

Actuation $\mathbf{d}$ is an explicit algebraic function of the state and the input. The direction along which information about R arrives is therefore not a fixed property of the instrumentation: it moves when the operating point moves. That is the observation this paper is built on, and Fig. 1 places it: the operating point reaches $\hat{\mathbf{d}}$ twice, once through the state and once directly.

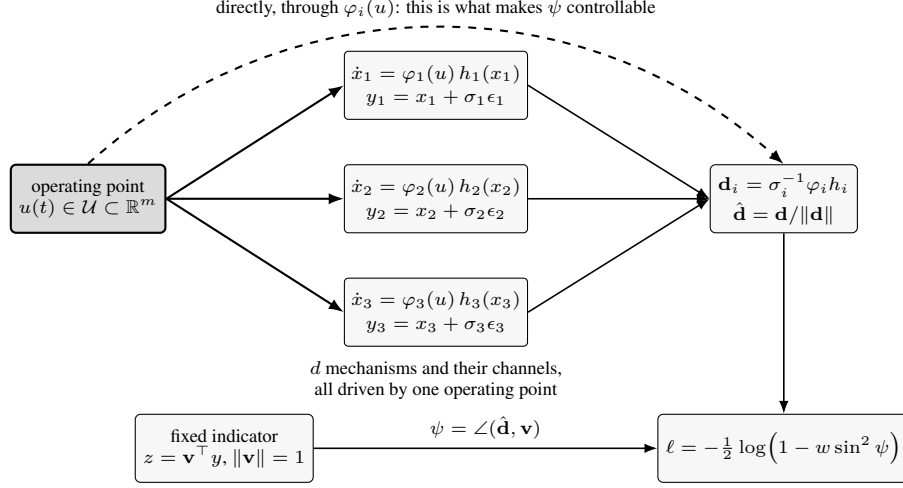
Because R is scalar, a single signal-to-noise ratio governs the information the record carries about it, namely the squared sensitivity times the prior uncertainty,

$$\gamma = \|\mathbf{d}\|^2 \text{Var}[R]. \quad (8)$$

Rates in (7) are taken with respect to clock time, whereas [3] differentiates with respect to age normalised by T_f . The two conventions differ by a factor common to all d channels, so they give the same $\hat{\mathbf{d}}$, and hence the same geometry in Sections 3–4, while rescaling γ . Every computation below uses clock time, including those of Section 5, where lifetime enters the objective.

2.4 The fixed health indicator and its information loss

A health indicator compresses the d measured channels into one scalar before any life prediction is made.



Alignment asks $\hat{d}(x, u) = q$: $d - 1$ scalar equations on m unknowns, whence the counting rule of Theorem 4.1.

Figure 1: The construction. The operating point enters twice, through the plant and directly through $\varphi_i(u)$; the second path is what makes the misalignment ψ a control variable. One input drives all d mechanisms.

We take it to be a fixed linear map. Since (7) already measures each channel in units of its own noise, we write the indicator in those same coordinates: with $\Sigma := \text{diag}(\sigma_1^2, \dots, \sigma_d^2)$ and $\tilde{y} := \Sigma^{-1/2} y$ the whitened record,

$$z = \mathbf{v}^T \tilde{y}, \quad \|\mathbf{v}\| = 1, \quad (9)$$

with the weight vector $\mathbf{v}$ chosen once, at design time, and held for the whole life of the component. Fixing it is not a modelling convenience: the map is typically embedded in firmware or frozen by certification, so it cannot be re-tuned as the component ages. An indicator specified on the raw measurements, $z = \mathbf{w}^T y$, the same map under $\mathbf{v} \propto \Sigma^{1/2} \mathbf{w}$, and the normalisation $\|\mathbf{v}\| = 1$ only fixes the scale of z ; it restricts nothing. The distinction matters because the angle below is measured after whitening: unless the σ_i are equal, a unit vector in raw coordinates is not a unit vector in whitened ones, and the two readings of ψ disagree.

We compute the cost of fixing $\mathbf{v}$ as follows. Write $\psi(t)$ for the angle between the informative direction $\hat{d}(t)$ and the indicator weights $\mathbf{v}$. After whitening, the signal in the record points along $\hat{d}$ and has magnitude $\|\hat{d}\| \sqrt{\text{Var}[R]}$, while the noise is isotropic of unit scale. Projecting onto a unit vector $\mathbf{v}$ therefore keeps the fraction $\cos \psi$ of the signal and all of the noise, so the indicator's signal-to-noise ratio is

$$\gamma \cos^2 \psi \quad (10)$$

against the γ of (8) available to an oracle that is free to re-aim its projection at $\hat{d}(t)$ at every instant. Since a scalar Gaussian channel of signal-to-noise ratio s carries $\frac{1}{2} \log(1 + s)$ nats about R [26], the information the fixed indicator forfeits, at each instant, is

$$\ell = \frac{1}{2} [\log(1 + \gamma) - \log(1 + \gamma \cos^2 \psi)] \geq 0, \quad (11)$$

The design objective throughout is its average over the life of the component,

$$\bar{\ell} := \frac{1}{T_f} \int_0^{T_f} \ell(t) dt, \quad (12)$$

a loss per unit time. We optimise this rate rather than the information the whole observation record forfeits, because (12) is unaffected by how densely the record is sampled and is therefore a property of the indicator rather than of the measurement schedule. The two are different quantities, and Proposition 2.3 gives the second in closed form.

A third quantity is the one an operator actually pays for: the error of whatever remaining-life estimator runs downstream. It is neither of the first two and is not a function of them. The loss is a logarithmic information quantity and the error is a time, and an estimator holding a whole history recovers much of what any single instant discards, so a large spread in (12) does not imply a comparable spread in error. What the theory can offer downstream is therefore an ordering rather than a conversion, and Section 6.8 tests that ordering against an estimator built from none of this construction: it survives, with a spread in error much smaller than the spread in loss. Equation (12) should be read accordingly, as an information-geometric objective whose ranking carries downstream, not as a surrogate for prognostic accuracy.

Two remarks on what (11) does and does not say. It is a loss relative to an oracle that no fixed map can realise, so it measures the price of the time-invariance constraint rather than of a bad design; a perfectly chosen $\mathbf{v}$ still pays it whenever $\hat{d}$ moves. And it depends on the indicator only through the single angle ψ : two indicators that make the same angle with $\hat{d}$ are equally good, whatever else distinguishes them.

The next lemma replaces (11) by a closed form that will be more convenient throughout, and records how it behaves at small misalignment.

Lemma 2.2. *Let $w := \gamma/(1 + \gamma) \in (0, 1)$. Then*

$$\ell = -\frac{1}{2} \log(1 - w \sin^2 \psi), \quad (13)$$

exactly and at every signal-to-noise ratio, and

$$\ell = \frac{w}{2} \psi^2 + \frac{w(3w-2)}{12} \psi^4 + O(\psi^6). \quad (14)$$

Proof. Substituting $\cos^2 \psi = 1 - \sin^2 \psi$ into (11) gives

$$\ell = -\frac{1}{2} \log \frac{1 + \gamma - \gamma \sin^2 \psi}{1 + \gamma} = -\frac{1}{2} \log(1 - w \sin^2 \psi).$$

For (14), use $-\frac{1}{2} \log(1 - s) = \frac{s}{2} + \frac{s^2}{4} + O(s^3)$ with $s = w \sin^2 \psi$ and $\sin^2 \psi = \psi^2 - \psi^4/3 + O(\psi^6)$; collecting the ψ^4 terms gives $-w/6 + w^2/4 = w(3w-2)/12$.

The quadratic term of (14) is the bound obtained in [3, Thm. 1], which is therefore the leading term of an exact identity rather than an approximation of unknown quality. The quartic coefficient $w(3w-2)/12$ changes sign at $w = 2/3$, that is at $\gamma = 2$: below that signal-to-noise ratio the quadratic model overstates the loss, above it the model understates it. This matters in practice because prognostic problems sit on both sides of $\gamma = 2$: an early-life measurement with a weak signal on one side, a well-instrumented component near failure on the other. Fig. 2(a) plots the exact loss against its quadratic approximation on both sides of that ratio.

Proposition 2.3. *Let the record consist of measurements (5) at instants $t_1, \dots, t_N$, with the noise independent across channels and across instants, and write $\mathbf{d}_k := \mathbf{d}(t_k)$ and ψ_k for the angle between $\hat{\mathbf{d}}_k$ and $\mathbf{v}$. Put*

$$G := \text{Var}[R] \sum_{k=1}^N \|\mathbf{d}_k\|^2, \quad W := \frac{G}{1 + G},$$

$$S := \frac{\sum_k \|\mathbf{d}_k\|^2 \sin^2 \psi_k}{\sum_k \|\mathbf{d}_k\|^2}.$$

Then the information about R that the fixed indicator forfeits over the whole record is

$$\ell_{\text{rec}} = -\frac{1}{2} \log(1 - WS). \quad (15)$$

Proof. See Supplementary Section S2.

Equation (15) is (13) again, with the instantaneous signal-to-noise ratio γ replaced by its sum G over the record, and $\sin^2 \psi$ by the mean S weighted by the signal energy $\|\mathbf{d}_k\|^2$. The reduction to one angle therefore survives the passage from an instant to a record. What does not survive is the averaging: $\bar{\ell}$ averages the loss, whereas (15) averages $\sin^2 \psi$ inside it, and the two differ. They also select different indicators. Minimising (15) over $\|\mathbf{v}\| = 1$ maximises $\mathbf{v}^\top (\sum_k \mathbf{d}_k \mathbf{d}_k^\top) \mathbf{v}$, so the

best fixed indicator for the record is the leading eigenvector of the integrated Fisher matrix, while the minimiser of $\bar{\ell}$ is the barycentre of Proposition 3.3. Every result below is stated for $\bar{\ell}$.

Corollary 2.4 (what the loss costs an estimator). *In the model of Section 2, the minimum-variance estimate of R formed from the fixed indicator carries mean-squared error*

$$\frac{\text{Var}[R | z]}{\text{Var}[R | \tilde{y}]} = \frac{1 + \gamma}{1 + \gamma \cos^2 \psi} = e^{2\ell}$$

times that of the estimate formed from the whole whitened record, at each instant. The same identity holds over a record with ℓ replaced by ℓ_{rec} of Proposition 2.3.

Proof. Write $\tilde{y} = \mathbf{d}R + \epsilon$ with $\epsilon \sim N(0, I_d)$ and R Gaussian. The indicator returns $z = (\mathbf{v}^\top \mathbf{d})R + \mathbf{v}^\top \epsilon$, whose noise is standard normal because $\|\mathbf{v}\| = 1$, so its signal-to-noise ratio is $\gamma \cos^2 \psi$ and $\text{Var}[R | z] = \text{Var}[R]/(1 + \gamma \cos^2 \psi)$. The record's sufficient statistic is $\mathbf{d}^\top \tilde{y}$, with ratio γ , giving $\text{Var}[R | \tilde{y}] = \text{Var}[R]/(1 + \gamma)$. Their quotient is $e^{2\ell}$ by (11). The record statement is the same computation with G and S in place of γ and $\sin^2 \psi$. $\square$

The loss is therefore not a score standing in for estimation error: it is the logarithm of the factor by which that error inflates. What the corollary does not settle is whether the ordering survives an estimator other than the one the model prescribes, and Section 6.8 checks that separately.

2.5 The control problems

The plant (2) is a nonlinear system driven by the operating point, and the fixed indicator is a static output functional of its measurements. The paper studies three problems on it, stated here so that each result below can be placed.

Problem 1 (attainability). Given the plant (2), the envelope $\mathcal{U}$ and an initial state x_0 , characterise the indicators and admissible inputs for which $\bar{\ell} = 0$, and count the independent inputs this requires.

Problem 2 (best achievable loss). For a fixed indicator, find the infimum of $\bar{\ell}$ over admissible inputs, the horizon T_f being determined by the input, and certify it.

Problem 3 (screening before design). From the plant model and $\mathcal{U}$ alone, before any indicator is chosen, decide whether steering can reduce $\bar{\ell}$ substantially below its value under the best constant input.

Theorem 4.1 answers Problem 1 exactly, as membership of the reachable direction set at every point of one segment that the indicator fixes, and gives the count of inputs as a necessary condition for generic data; Corollary 4.3 separates it from the weaker task of holding the misalignment constant, and Proposition 4.5 restates the

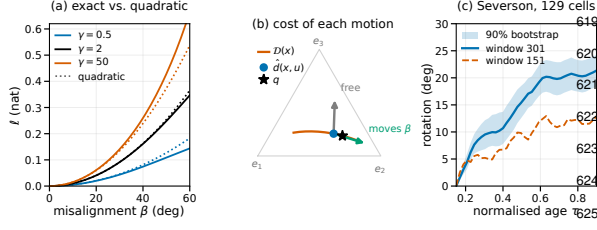


Figure 2: (a) Lemma 2.2: the exact loss (solid) against its quadratic model (dotted), whose quartic coefficient changes sign at $\gamma = 2$. (b) Theorem 3.6: of the replicator velocity, only the component along the geodesic towards q moves β . (c) The rotation, measured on the Severson fleet at two differentiation windows.

exact criterion on the separable subclass. Problem 2 is the fractional optimal control problem of Section 5. Its value is bracketed numerically in Section 6.6, and Proposition 5.1 reduces its certification to exhibiting one verification function, which remains open; Section 6.9 measures how much of the computed benefit survives an error in the model the design is built on. Problem 3 is addressed empirically, across random plants of the separable model, in Section 6.7. Proposition 4.7 shows that the objective of Problems 1 and 2 cannot be read from the record's Fisher information about remaining life, only from its ratio to the indicator's own, so neither problem reduces to input design that scores the record.

3 The information simplex and its replicator flow

Lemma 2.2 reduced the loss to one angle. This section shows that the angle lives on a familiar object, the probability simplex, and that the degradation traces a flow along it that the operating point drives. The payoff is threefold: the loss becomes a recognised statistical distance, the optimal indicator becomes a barycentre, and the effect of the input becomes a single interpretable quantity. Two of the results here are load-bearing for what follows — the replicator form of Theorem 3.4 and the misalignment rate of Theorem 3.6, which Section 4 builds the reachability argument on. The Bhattacharyya identity and the barycentre are interpretation: they say what the loss and the optimal indicator already are in recognised terms, and nothing later depends on them.

3.1 Information shares

Because every entry of $\mathbf{d}$ is positive (Section 2.3), the squared components of the unit vector $\hat{\mathbf{d}}$ form a probability vector:

$$p_i := \hat{\mathbf{d}}_i^2, \quad p_i > 0, \quad \sum_i p_i = 1. \quad (16)$$

The interpretation is direct. The total Fisher information the record carries about R is $\|\mathbf{d}\|^2 \text{Var}[R]$ by (8), and

channel i contributes $\mathbf{d}_i^2 \text{Var}[R]$ of it. So p_i is exactly the share of the total information about remaining life that arrives through channel i . A component in which one mechanism dominates the degradation has p near a vertex of the simplex; one in which three mechanisms contribute equally sits near its centre.

The same map applied to the indicator, $q_i := \mathbf{v}_i^2$, is legitimate because an optimal indicator may be taken with non-negative weights.

Remark 3.1 (positivity is load-bearing). *If $\hat{\mathbf{d}} \geq 0$ then $\hat{\mathbf{d}}^\top |\mathbf{v}| \geq |\hat{\mathbf{d}}^\top \mathbf{v}|$ at every instant, so replacing $\mathbf{v}$ by $|\mathbf{v}|$ never increases (11): non-negative weights are without loss of generality. This argument uses the positivity of $\mathbf{d}$ inherited from $\varphi_i, h_i > 0$, and it fails without it: in random trials with sign-indefinite $\hat{\mathbf{d}}$, taking absolute values was worse than the signed vector about a third of the time. The results therefore require the positive-orthant structure of the degradation model; it is not only a notational convenience.*

3.2 The loss as a Bhattacharyya distance

The identification below is used twice in what follows: it gives “the average of a curve on the simplex” an unambiguous meaning (Section 3.3), and it lets the high-signal limit of the loss be read off as a known divergence.

Proposition 3.2. $\cos \psi = \sum_i \sqrt{p_i q_i}$ is the Bhattacharyya coefficient of p and q . Consequently $\psi = \frac{1}{2} d_{\text{FR}}(p, q)$, where d_{FR} is the Fisher–Rao distance on the simplex, and, writing $D_{\text{B}}(p, q) = -\log \sum_i \sqrt{p_i q_i}$ for the Bhattacharyya divergence,

$$\ell = -\frac{1}{2} \log[(1 - w) + w e^{-2D_{\text{B}}(p, q)}]. \quad (17)$$

As $\gamma \rightarrow \infty$, $\ell \rightarrow D_{\text{B}}(p, q)$.

Proof. By construction,

$$\cos \psi = \hat{\mathbf{d}}^\top \mathbf{v} = \sum_i \sqrt{p_i} \sqrt{q_i},$$

which is precisely the Bhattacharyya coefficient of p and q . The map $p \mapsto \sqrt{p}$ is the classical isometry between the simplex under the Fisher–Rao metric and the positive orthant of the unit sphere, under which the geodesic distance is twice the spherical angle [21, 22]; hence $\psi = \frac{1}{2} d_{\text{FR}}$. Then $\cos^2 \psi = e^{-2D_{\text{B}}}$, and substituting into (13) gives (17). Letting $w \rightarrow 1$ gives the limit. $\square$

The square root is a bijection between the simplex and the non-negative part of the unit sphere, carrying p to $\hat{\mathbf{d}}$ and q to $\mathbf{v}$, and by the proposition it is an isometry. We use the two descriptions interchangeably from here on: a statement such as $q \in \mathcal{D}(x)$ in Section 4 says that $\mathbf{v}$ is a reachable direction, and a distance between points of the simplex is the angle between the corresponding unit vectors.

The resulting picture is the one the rest of the paper develops. The degrading component is a curve on the simplex, the health indicator is a single point on it, and the information lost depends only on the distance between them. Design of the indicator is the choice of one point; control of the operating point is the shaping of the curve.

3.3 The optimal indicator as a barycentre

If the indicator is one point and the plant is a curve, the best point should be some kind of average of the curve. It is, but of two different kinds in two different limits.

Proposition 3.3. *Let q^* minimise the time-averaged loss.*

- (i) *As $\gamma \rightarrow \infty$, $\ell \rightarrow D_B$ pointwise, q^* tends to the Bhattacharyya barycentre $\arg \min_q \int D_B(p_\tau, q) d\tau$.*
- (ii) *As the curve shortens, $\ell = \frac{w}{2}\psi^2 + O(\psi^4)$, so q^* tends to the w -weighted Fisher–Rao barycentre, and $\bar{\ell} = \frac{1}{8}w \text{Var}^{\text{FR}} + O(\psi^4)$ where $\text{Var}^{\text{FR}} = \min_q \mathbb{E}_\mu[d_{\text{FR}}^2(p_\tau, q)]$ is the μ -weighted Fréchet variance of the curve.*

Proof. See Supplementary Section S2.

The two limits are taken in different parameters, signal-to-noise ratio in (i) and arc length in (ii), and they do not coincide. On a curve of angular span 82.7° the two barycentres lie 0.72° apart, and q^* crosses from the Fisher–Rao side to the Bhattacharyya side between 0 and 10 dB, reaching 3×10^{-5} degrees of the latter by 50 dB. For design purposes the practical consequence is mild: either barycentre of a nominal degradation path is a good starting indicator.

3.4 The replicator flow and the role of the input

We now ask how fast the informative direction moves and what the operating point does to that motion.

Theorem 3.4. *Let $g_i = \log \mathbf{d}_i$ denote the logarithmic rate of channel i . Along (2),*

$$\dot{\mathbf{d}}_i = \hat{\mathbf{d}}_i(\dot{g}_i - \mathbb{E}_p[\dot{g}]), \quad (18)$$

so the information shares obey the replicator equations of evolutionary dynamics, in which each share grows in proportion to the amount by which its own rate exceeds the average, the average being taken over the shares themselves:

$$\dot{p}_i = 2p_i(\dot{g}_i - \mathbb{E}_p[\dot{g}]), \quad (19)$$

and

$$\|\dot{\mathbf{d}}\| = \text{Std}_p[\dot{g}], \quad \dot{g} = [\varphi_i(u)h'_i(x_i)]_i + \nabla_u \log \varphi \quad (20)$$

Proof. From $\mathbf{d}_i = e^{g_i}$ we have $\dot{\mathbf{d}}_i = \mathbf{d}_i \dot{g}_i$, so $\dot{\mathbf{d}} = \|\mathbf{d}\|^{-1}(\dot{\mathbf{d}} - \hat{\mathbf{d}}(\hat{\mathbf{d}}^\top \dot{\mathbf{d}}))$ has components $\dot{\mathbf{d}}_i \dot{g}_i - \hat{\mathbf{d}}_i \sum_j \hat{\mathbf{d}}_j^2 \dot{g}_j$, which is (18); squaring and summing gives $\text{Var}_p[\dot{g}]$. Since $p_i = \hat{\mathbf{d}}_i^2$, $\dot{p}_i = 2\hat{\mathbf{d}}_i \dot{\mathbf{d}}_i = 2p_i(\dot{g}_i - \mathbb{E}_p[\dot{g}])$. Differentiating $g_i = \log \varphi_i(u) + \log h_i(x_i) - \log \sigma_i$ along (2) gives $\dot{g}_i = \nabla_u \log \varphi_i \cdot \dot{u} + (h'_i/h_i)\varphi_i \dot{h}_i$, and $(h'_i/h_i)\varphi_i \dot{h}_i = \varphi_i \dot{h}'_i$. $\square$

Equation (20) has a compact reading. The speed at which the informative direction rotates equals the information-weighted spread of the mechanisms' logarithmic rates. If all mechanisms accelerate at the same proportional rate the direction does not move at all, however fast the component is degrading; it is disagreement among the mechanisms that rotates it. The identity is the Fisher–Rao/Shahshahani geometry of the simplex combined with Fisher's fundamental theorem of natural selection [21, 23, 24]. What is new here is its identification with the informative direction of a degrading component, and, decisively, that the quantity playing the role of fitness, the logarithmic rate $\dot{g}_i$ of channel i , is actuated: the second summand in (20) is the operator's lever.

Remark 3.5 (contrast with Oja's flow). *Oja's flow for online principal-component extraction has the same algebraic form; for a diagonal covariance it reduces to a replicator equation with fitness twice the eigenvalues [27]. The roles are opposite. There the flow is an algorithm, run to fixation in order to find a direction, with constant fitness. Here it is the plant, its fitness is time-varying and actuated, and the aim is to prevent fixation from carrying the direction away from a fixed indicator.*

Only one scalar matters for the loss. Fix the indicator q and write $\beta := \angle(\hat{\mathbf{d}}, \mathbf{v}) = \frac{1}{2}d_{\text{FR}}(p, q)$ for the misalignment. This is the angle ψ of Lemma 2.2 under a new name, and the change of name marks a change of role: ψ was a fixed angle between two directions, whereas $\beta = \beta(x_t, u_t)$ is followed along the trajectory and is the quantity the input acts on. Every loss expression below may therefore be read as a function of β .

Theorem 3.6. *Let the operating-point trajectory $u(\cdot)$ be absolutely continuous, and let $\hat{\mathbf{e}}$ be the unit tangent at $\hat{\mathbf{d}}$ along the geodesic towards $\mathbf{v}$. Then $\dot{\beta} = -\langle \dot{\hat{\mathbf{d}}}, \hat{\mathbf{e}} \rangle$, which is affine in the input rate: $\dot{\beta} = \alpha_0(x, u) + \alpha(x, u)^\top \dot{u}$. In particular $|\dot{\beta}| \leq \|\dot{\hat{\mathbf{d}}}\|$, with equality only for motion along that geodesic.*

Proof. $\cos \beta = \hat{\mathbf{d}}^\top \mathbf{v}$ gives $-\sin \beta \dot{\beta} = \dot{\hat{\mathbf{d}}}^\top \mathbf{v}$. Also $\hat{\mathbf{e}} = (\mathbf{v} - \cos \beta \hat{\mathbf{d}})/\sin \beta$ and $\hat{\mathbf{d}}^\top \hat{\mathbf{d}} = 0$, so $\langle \dot{\hat{\mathbf{d}}}, \hat{\mathbf{e}} \rangle = \dot{\hat{\mathbf{d}}}^\top \mathbf{v}/\sin \beta = -\dot{\beta}$. Affinity follows because $\dot{g}$ is affine in $\dot{u}$ by (20) and $\dot{\hat{\mathbf{d}}}$ is linear in $\dot{g}$. $\square$

The hypothesis is needed only to write $\dot{u}$; nothing in Section 4 depends on it, since the reachable sets (21) are defined pointwise in u and the limits below hold for

every measurable admissible input. Theorem 3.6 is a local kinematic description of smooth steering, not a restriction on the controls the rest of the paper admits. The component of the replicator velocity orthogonal to $\hat{e}$ is therefore free: the informative direction may rotate in a $(d-2)$ -dimensional subspace at no cost in information. Fig. 2(b) shows the decomposition at one state. This is the observation that separates the two objectives of the next section.

4 Reachability limits on the misalignment

Controlling the rate $\dot{\beta}$ is not the same as reaching a value of β . The distinction is the subject of this section, and it is what decides whether a fixed indicator can be made lossless at all.

For a state x , the set of informative directions the operator can produce by choosing the operating point is

$$\mathcal{D}(x) := \{\hat{\mathbf{d}}(x, u) : u \in \mathcal{U}\} \subset S^{d-1}, \quad (21)$$

the reachable direction set, and the closest the indicator can be brought at that state is the reachable misalignment floor

$$\beta_{\min}(x, q) := \text{dist}(q, \mathcal{D}(x)). \quad (22)$$

For a single input, $\mathcal{D}(x)$ is an arc traced out as u sweeps $\mathcal{U}$; its length measures the actuation authority available at x , and it shrinks to a point when the mechanisms respond identically to the input.

Zero loss leaves the trajectory almost no freedom. Once $\hat{\mathbf{d}}$ is pinned to $\mathbf{v}$, the state can only advance along one fixed direction, because $\dot{x} = \Sigma^{1/2} \mathbf{d} = \|\mathbf{d}\| \Sigma^{1/2} \mathbf{v}$ by (7), and the input is left to choose only the speed. Write

$$\mathcal{S}_q := \{x_0 + s \Sigma^{1/2} \mathbf{v} : 0 \leq s \leq s_f\} \quad (23)$$

for the segment from x_0 in that direction, stopped at the value s_f at which it meets the condemnation surface $\{\max_i x_i = x_f\}$ of (3).

Theorem 4.1. Fix an indicator $\mathbf{v}$, with share vector $\mathbf{q}$, and let each h_i be locally Lipschitz.

- (i) An admissible input attains $\bar{\ell} = 0$ only if its trajectory runs along $\mathcal{S}_q$, and zero loss is attainable with q if and only if $q \in \mathcal{D}(x)$ for every $x \in \mathcal{S}_q$.
- (ii) If $m < d - 1$ and $u \mapsto \hat{\mathbf{d}}(x_0, u)$ is continuously differentiable, the indicators that attain zero loss form a null subset of the unit sphere, which almost every q misses.
- (iii) Along any admissible trajectory, $\bar{\ell} \geq \frac{1}{T_f} \int_0^{T_f} \ell(\beta_{\min}(x_t, q)) dt$, with the weight w of Lemma 2.2 evaluated at (x_t, u_t) .

Proof. (i) By Lemma 2.2, $\bar{\ell} = 0$ iff $\sin \beta = 0$, so $\bar{\ell} = 0$ iff $\hat{\mathbf{d}}(x_t, u_t) = \mathbf{v}$ for almost every t . Then $\dot{x}_t = \|\mathbf{d}_t\| \Sigma^{1/2} \mathbf{v}$ almost everywhere, so $x_t = x_0 + s(t) \Sigma^{1/2} \mathbf{v}$ with $s(t) = \int_0^t \|\mathbf{d}_\tau\| d\tau$ continuous and strictly increasing: the trajectory runs along $\mathcal{S}_q$ and fails where $\mathcal{S}_q$ meets the condemnation surface. As $q \in \mathcal{D}(x_t)$ for almost every t , q lies in $\mathcal{D}(x)$ for x in a dense subset of $\mathcal{S}_q$. The set $\{x : q \in \mathcal{D}(x)\}$ is closed, since $\mathcal{U}$ is compact and $\hat{\mathbf{d}}$ continuous, so it contains all of $\mathcal{S}_q$. Conversely, let $q \in \mathcal{D}(x)$ for every $x \in \mathcal{S}_q$. The map $s \mapsto \{u \in \mathcal{U} : \hat{\mathbf{d}}(x_0 + s \Sigma^{1/2} \mathbf{v}, u) = \mathbf{v}\}$ has non-empty compact values and a closed graph, so it admits a measurable selection $\bar{u}(s)$ [28, Thm. 8.1.3]. On the compact set $\mathcal{S}_q \times \mathcal{U}$ the norm $\|\mathbf{d}\|$ is continuous and positive, hence bounded below, so $t(s) = \int_0^s \|\mathbf{d}(x_0 + \sigma \Sigma^{1/2} \mathbf{v}, \bar{u}(\sigma))\|^{-1} d\sigma$ is finite, absolutely continuous and strictly increasing. With $s(t)$ its inverse, the input $u(t) = \bar{u}(s(t))$ is measurable, and $x(t) = x_0 + s(t) \Sigma^{1/2} \mathbf{v}$ satisfies $\dot{x} = \|\mathbf{d}\| \Sigma^{1/2} \mathbf{v} = \Sigma^{1/2} \mathbf{d}(x, u)$, which is (2), almost everywhere. Because h is locally Lipschitz, this is the trajectory the input generates. It reaches the condemnation surface at the finite time $t(s_f)$ with $\beta = 0$ throughout. (ii) By (i), zero loss needs $q \in \mathcal{D}(x_0)$. For $m < d - 1$ that set, the image of the m -dimensional compact set $\mathcal{U}$ under a continuously differentiable map, is a null subset of the $(d - 1)$ -dimensional sphere, which almost every q misses. (iii) $\beta_t \geq \beta_{\min}(x_t, q)$ for every admissible input by definition of $\beta_{\min}$ as a minimum over $\mathcal{U}$, and $\bar{\ell}$ is increasing in β on $[0, \pi/2]$ at fixed w . $\square$

The condition in (i) has to be read along $\mathcal{S}_q$ and not along the trajectory of an input one happens to hold. An aligning input generates its own trajectory, and reachability along any other trajectory says nothing.

Remark 4.2 (reachable along a trajectory, unattainable on the segment). Take $d = 2$, $m = 1$, unit noise, $\varphi_i(u) = e^{\theta_i - E_i u}$ with $\theta = (0, 0.5)$, $E = (1, 2)$ and $\mathcal{U} = [0, 1.3]$, shapes $h_1 = 1$ and $h_2(x) = e^x$, $x_0 = 0$, $x_f = 0.9$, and the indicator with $\log(\mathbf{v}_2/\mathbf{v}_1) = -0.2$. It is aligned at state x by the input $u = 0.7 + x_2$, which is admissible exactly when $x_2 \leq 0.6$. Under the constant input $u = 1.3$ the plant fails with $x_2 = 0.52$, so the indicator is reachable at every state that trajectory visits. Zero loss is nevertheless unattainable: along $\mathcal{S}_q$, $x_2 = e^{-0.2} x_1$, and x_2 reaches 0.6 at $x_1 = 0.73$, before failure.

The bound (iii) is state-conditioned. No input violates it, but it holds along the trajectory the input has chosen, and that trajectory is itself steered. Optimising over trajectories is deferred to Section 5. The two levels are the structure of the argument: a state-conditioned limit here, a trajectory-level limit there.

The practical reading is a counting rule, and its parts have different standing. The measure statement is proved: with fewer than $d - 1$ independently adjustable operating variables, almost every indicator already fails at x_0 , by (ii). That no indicator at all can then

be held is a count of dimensions rather than a theorem. Along $\mathcal{S}_q$, membership of q in a reachable set of dimension m inside the $(d-1)$ -dimensional sphere imposes $d-1-m$ conditions at every point of a segment. Generic plant data do not meet them, and Remark 4.6 constructs plants that do, which we do not prove rare. Conversely, $m \geq d-1$ is not sufficient. It lets the aligning equation be solved locally, but zero loss also needs the aligning input to stay inside $\mathcal{U}$ along the whole of $\mathcal{S}_q$. That envelope condition is part (i) itself and has to be checked on the segment. Section 6.4 checks it for a plant with two inputs. For a generic plant under this model, three mechanisms and a single temperature setpoint will not do it. Proposition 4.5 states the condition exactly on the separable subclass. But a weaker and much cheaper objective is available.

Corollary 4.3 (what actuation is actually required). *Two objectives that are easily conflated carry different requirements.*

(i) *Driving $\bar{\ell}$ to zero is equivalent to freezing $\hat{\mathbf{d}}$ at q , and needs $d-1$ conditions on $\dot{u}$, hence generically $m \geq d-1$.*

(ii) *Holding β at a constant level, not necessarily zero, is the single scalar equation $\alpha_0 + \alpha^\top \dot{u} = 0$, solvable whenever $\alpha \neq 0$ and the resulting $u(\cdot)$ stays in $\mathcal{U}$; generically $m \geq 1$ suffices, at least locally in time.*

Proof. (i) By Theorem 4.1, $\bar{\ell} = 0$ iff $\hat{\mathbf{d}}(x_t, u_t) = q$ for almost every t , so $\hat{\mathbf{d}}$ is constant; by Theorem 3.4 that forces $\dot{g}_i = \mathbb{E}_p[\dot{g}]$ for every i , which is $d-1$ independent conditions on $\dot{u} \in \mathbb{R}^m$. (ii) is one equation in $\dot{u}$ by Theorem 3.6. The gap between them is the content of that theorem: β sees only the component of $\hat{\mathbf{d}}$ along $\hat{e}$, so the remaining $d-2$ directions of rotation are unconstrained by (ii) and constrained by (i).

Minimising $\bar{\ell}$ is neither of these: it trades the level of β against the trajectory that carries it, and is the problem of Section 5.

4.1 The reachable sets are translates of one set

Theorem 4.1 counts equations against unknowns and hedges with “generic”. The hedge is not idle, and the following observation says exactly what it covers.

Lemma 4.4 (separability). *Write $\xi(x, u)$ for the class of $\log \mathbf{d}(x, u)$ modulo the all-ones vector, which is what $\mathbf{d}$ retains. Then*

$$\xi(x, u) = a(x) + b(u),$$

$$a_i(x) = \log \frac{h_i(x_i)}{\sigma_i}, \quad b_i(u) = \log \varphi_i(u), \quad (24)$$

both taken modulo the all-ones vector. Consequently every reachable direction set is the same set translated. $\mathcal{D}(x) = a(x) + \mathcal{B}$ with $\mathcal{B} := b(\mathcal{U})$ fixed.

Proof. Immediate from $\mathbf{d}_i = \sigma_i^{-1} \varphi_i(u) h_i(x_i)$ in (7): the logarithm turns the product into a sum, and the normalisation of $\hat{\mathbf{d}}$ removes any component along the all-ones vector. $\square$

The decomposition separates the two parties. The set $\mathcal{B}$ is fixed by the stress model φ and the envelope $\mathcal{U}$ alone (Fig. 3(a)). An operator’s authority can therefore be computed offline from the accelerated-life model, with no degradation record at all. The curve $\mathcal{A} := \{a(x_t)\}$ is fixed by the damage shapes h and the noise levels σ , and every reachable set is that one set carried along it (Fig. 3(b)). What the geometry of Section 3 measures is how far the plant’s demand $\mathcal{A}$ sticks out of the operator’s authority $\mathcal{B}$.

That offline route is the most useful consequence drawn in this paper and also the most restrictive. It holds on the separable subclass and not beyond it, and the best high-fidelity turbofan data we have reject it jointly with the channel pairing of (5) (Section 7). This subsection and Proposition 4.5 should therefore be read as describing a structured subclass that is convenient when it applies, not as a general account of real machinery. Fig. 6 marks the one branch of the development that depends on it.

Proposition 4.5 (exact criterion). *Write κ for the quotient class of $\log \mathbf{v}$ and $\mathcal{A}_q := a(\mathcal{S}_q)$ for the image of the segment (23). Zero loss is attainable with q if and only if $\mathcal{A}_q \subseteq \kappa - \mathcal{B}$. In particular the curve $\mathcal{A}_q$ must lie in a translate of $-\mathcal{B}$, a set of dimension at most m , which for a curve in the $(d-1)$ -dimensional quotient fails generically unless $m \geq d-1$. When it does not fail, $m < d-1$ suffices.*

Proof. By Theorem 4.1(i), zero loss is attainable iff $q \in \mathcal{D}(x)$ for every $x \in \mathcal{S}_q$. By Lemma 4.4 that reads $\kappa - a(x) \in \mathcal{B}$ for every $x \in \mathcal{S}_q$, that is $\mathcal{A}_q \subseteq \kappa - \mathcal{B}$. The dimension statement holds because $\mathcal{B} = b(\mathcal{U})$ is the image of an m -dimensional set; when b is affine in u , as for Arrhenius factors written in reciprocal temperature, it spans an affine subspace of dimension at most m . $\square$

The containment is drawn in Fig. 3(c), and the mechanism the count obscures is now visible. The indicator fixes the curve $\mathcal{A}_q$ before any input is chosen. Alignment at a state then fixes the input outright, since $b(u) = \kappa - a(x)$ determines it, and the only question left is whether that input is admissible all the way along. One input can carry the whole burden only when $\mathcal{A}_q$ already lies along $\mathcal{B}$.

Remark 4.6 (a family with $m = 1$ and any d). *Take exponential shapes $h_i(x) = e^{\nu_i(x_i - x_i^0)}$, which (2) admits, and Arrhenius factors written in reciprocal temperature, $\varphi_i(u) = e^{\theta_i - E_i u}$ with $u = 1/(kT)$, so that b is affine in u . Under the ramp $u(t) = u_0 + rt$ the damage rates are constant, $x_i(t) = x_i^0 + \varphi_i^0 t$ with $\varphi_i^0 = \varphi_i(u_0)$, so $a(x_t)$ moves at constant velocity $\nu_i \varphi_i^0$. Choosing $\nu_i = r E_i / \varphi_i^0$ makes that velocity parallel to E , which*

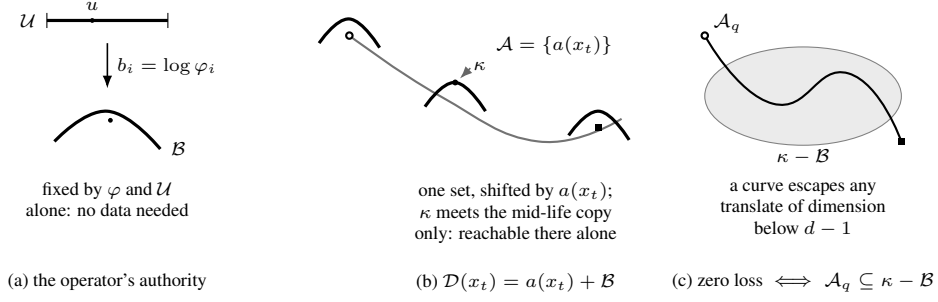


Figure 3: How the counting rule becomes a containment. (a) The operator's authority $\mathcal{B} = b(\mathcal{U})$, fixed by the stress model and the envelope alone. (b) Every reachable set is that one set translated, $\mathcal{D}(x_t) = a(x_t) + \mathcal{B}$ (circle: start of life, square: failure). (c) Zero loss asks the image $\mathcal{A}_q = a(\mathcal{S}_q)$ of the segment that the indicator fixes to lie in one translate (Proposition 4.5).

is the direction of $\mathcal{B}$, and $b(u_t)$ retreats along $\mathcal{B}$ at the same rate. The two cancel and $\hat{\mathbf{d}}$ is frozen. Supplementary Section S4 integrates the construction for $d = 2$.

This bears on the sharpness of Theorem 4.1. The family of Remark 4.6 is a knife edge. At one percent of error in its shape coefficients, steering is already worse than not steering (Supplementary Section S4). A designer should read Proposition 4.5 as an explanation of when the counting rule binds, and Theorem 4.1 as the rule to design against.

4.2 The loss and Fisher-information criteria

A natural objection is that all of this should follow from designing the input to maximise the Fisher information about R , as in the observability-aware literature discussed in Section 1. Two Fisher informations are in play, and they answer the objection differently. The record's, $I_{\text{rec}} := \|\mathbf{d}\|^2$ per unit prior variance, is what that literature scores. The indicator's own, $I_z := (\mathbf{v}^\top \mathbf{d})^2$, the information about R in the scalar z of (9).

Proposition 4.7 (the record's information does not fix the loss). *The loss of a fixed indicator is*

$$\ell = \frac{1}{2} \log(1 + \text{Var}[R] I_{\text{rec}}) - \frac{1}{2} \log(1 + \text{Var}[R] I_z), \quad (25)$$

so it is a function of the pair (I_{rec}, I_z) . It is not a function of I_{rec} alone: at any fixed value of I_{rec} , ℓ takes every value in $[0, \frac{1}{2} \log(1 + \gamma))$ as the misalignment ranges over $[0, \pi/2)$. No scalarisation of the record's Fisher information therefore determines the loss, and an input chosen to maximise one need not reduce it.

Proof. Equation (25) is (11) with $\gamma = \text{Var}[R] I_{\text{rec}}$ and $\gamma \cos^2 \psi = \text{Var}[R] (\mathbf{v}^\top \mathbf{d})^2$. For a scalar latent the record's Fisher information is the number I_{rec} , its trace, determinant and least eigenvalue coincide and every scalarisation is a strictly monotone function of it. Hold I_{rec} , and hence γ and the weight w of Lemma 2.12,

fixed. Then $\ell = -\frac{1}{2} \log(1 - w \sin^2 \psi)$ is continuous and strictly increasing in ψ on $[0, \pi/2)$, from 0 towards $\frac{1}{2} \log(1 + \gamma)$. $\square$

Stated in engineering terms, the record's information measures the strength of the signal, and the loss depends also on its direction relative to the indicator. The proposition is not a limitation of Fisher criteria as such. It says which one to use: I_z , compared with I_{rec} , carries the loss exactly, so observability-aware input design applies to a fixed indicator once it scores the indicator's information rather than the record's. Section 6 exhibits the difference: with the record's score held equal to seven digits, the loss still spans a factor of 9.4, and I_z orders the configurations as the loss does.

5 The best achievable loss as a fractional control problem

Theorem 4.1 bounds the loss along a given trajectory. But the input steers the trajectory as well as the misalignment on it, so the best achievable loss requires optimising over both at once. This changes the structure of the problem.

The objective is a time average over a lifetime that the input itself determines:

$$\bar{\ell}^*(q) = \min_{u(\cdot) \in \mathcal{U}} \frac{\int_0^{T_f} \ell(\beta(x_t, u_t)) dt}{T_f}. \quad (26)$$

Both the integrand and the upper limit T_f depend on $u(\cdot)$. This makes (26) a fractional optimal control problem, meaning that the objective is a ratio rather than that the dynamics carry a fractional-order derivative. The fraction is not a technicality: it is the source of the trade-off, as follows.

Were the horizon fixed, the input would only have to keep β small over a known interval. Because T_f is itself controlled, the denominator of (26) enters as well. The operating point that aligns the indicator is in general not the operating point that prolongs life, so aligning now

changes the interval over which the alignment is averaged. The optimum must trade the two. Section 6.5 measures how steep that trade is on a concrete plant, and Section 6.6 how much a policy that ignores it loses. A fixed-horizon formulation represents none of it, which is why one was tried and discarded (Section 6.6).

5.1 A converse bound uniform over trajectories

Before computing anything we record a certificate that removes the conditioning. Theorem 4.1 bounds the loss along a trajectory the input has already chosen; the following proposition bounds it over all of them at once, and it is what would make the number computed below a limit rather than an observation, were a function satisfying it exhibited; none is, for reasons given below.

Proposition 5.1 (verification certificate). *Fix an indicator q and a constant $\lambda \geq 0$. Suppose there is a continuously differentiable V on a neighbourhood of the admissible state region such that*

(i) $V(x) \leq 0$ for every admissible x on the condemnation surface $\{\max_i x_i = x_f\}$ of (3), and

(ii) writing $D_i(x, u) = \varphi_i(u)h_i(x_i)$ for the drift of (2), for every admissible state x and every $u \in \mathcal{U}$,

$$\ell(\beta(x, u)) - \lambda + \nabla V(x)^\top D(x, u) \geq 0. \quad (27)$$

Then every admissible input satisfies $\bar{\ell} \geq \lambda + V(x_0)/T_f$. In particular, if $V(x_0) \geq 0$ then $\bar{\ell}^*(q) \geq \lambda$, uniformly over all admissible inputs and hence over all trajectories.

Proof. Let $u(\cdot)$ be any admissible input and x_t its trajectory. Since V is C^1 and $t \mapsto x_t$ is absolutely continuous, $t \mapsto V(x_t)$ is absolutely continuous and the chain rule gives $\frac{d}{dt}V(x_t) = \nabla V(x_t)^\top \dot{x}_t = \nabla V(x_t)^\top D(x_t, u_t)$ for almost every t . Applying (27) at (x_t, u_t) gives $\ell(\beta(x_t, u_t)) - \lambda + \frac{d}{dt}V(x_t) \geq 0$ a.e., and integrating over $[0, T_f]$,

$$\int_0^{T_f} (\ell - \lambda) dt \geq V(x_0) - V(x_{T_f}) \geq V(x_0),$$

the last step by (i), since x_{T_f} lies on the condemnation surface by (3). Dividing by $T_f > 0$ gives the claim.

This is the standard verification argument for exit-time problems, applied to the Dinkelbach-transformed running cost $\ell - \lambda$ [29]; the proposition records it in the form the fractional problem needs. Condition (27) constrains the plant data alone: it quantifies over every state and every admissible input, and mentions no trajectory. Any V one can exhibit therefore certifies a limit, and the search for a converse becomes the search for a verification function.

Continuous differentiability is a convenience of the proof, not a requirement of the argument. A continuous

viscosity subsolution of the corresponding Hamilton–Jacobi–Bellman equation that satisfies (i) lies below the value function by the comparison principle, since the drift points out of the state region across the condemnation surface, and a Lipschitz V satisfying (27) with Clarke’s generalised gradient certifies in the same way [29]. The natural candidate is the value function of the Dinkelbach problem below, which satisfies (27) with equality in the viscosity sense. What is available in practice, however, is a grid approximation of it, and Supplementary Section S3 finds that candidate violating (27) on a finer grid by amounts that are discretisation error rather than a defect of the candidate. A certificate therefore needs either a discrete value function shifted by a bound on its own residual, or an error estimate for the monotone scheme that produced it. We do not construct one here, and the optimum computed below remains a computation; what Proposition 5.1 buys is the replacement of an open problem without shape by a concrete object with a checkable property.

Problems of the form (26) are handled by Dinkelbach’s transform [30]: setting

$$F(\lambda) = \min_{u(\cdot) \in \mathcal{U}} \int_0^{T_f} (\ell - \lambda) dt, \quad (28)$$

the map F is decreasing in λ and vanishes exactly at the optimal ratio, because every admissible lifetime lies between the shortest and the longest the envelope allows, both positive and finite. The inner problem in (28) has free terminal time and an absorbing set, hence a stationary Hamilton–Jacobi–Bellman equation on the state alone, with no time argument. We discretise it by a semi-Lagrangian scheme with multilinear interpolation [31], which is monotone and consistent, so its solutions converge to the viscosity solution as the grid is refined [31]. Because $\dot{x} > 0$, the interpolation cell of every foot point has all its corners other than the node itself at a larger index sum, so the discrete equation is solved exactly by one sweep in decreasing index sum, with one scalar equation per node for its own weight, and λ^* by Dinkelbach’s own iteration, which terminates because the discrete policies are finite. One implementation detail matters: the scheme must use a fixed spatial step rather than a fixed time step, because degradation speeds vary by orders of magnitude across $\mathcal{U}$ and a fixed time step is unusable at one end of the envelope or the other.

6 Numerical study

Sections 3–5 are analytical, and three of their claims are stated sharply enough to be tested rather than illustrated: that the record’s Fisher information does not fix the loss, that zero loss fails with one input and is attained with two when the envelope condition holds along the segment, and that the best achievable loss is a finite number one can bracket. This section instantiates the model on a

synthetic three-mechanism plant on which every quantity is available in closed form, checks those three claims against it, measures what steering is worth, asks which quantity, available in advance, says whether it will repay the effort, and measures how much of the benefit survives an error in the stress model. Section 7 then reports what measured fleets can and cannot establish about the premises.

6.1 The test plant

We take $d = 3$ mechanisms with Arrhenius factors $\varphi_i(u) = e^{\theta_i - E_i u}$, in which the input u is the reciprocal temperature $1/(kT)$, so that this is the Arrhenius form of Section 2 with e^{θ_i} the pre-exponential constant A_i . We set $\theta = (0, 0.3, -0.2)$ and $E = (0.6, 1.1, 0.8)$, and shapes $h_i(x) = 1 + c_i x$ with $c = (-0.5, 2.0, 0.8)$, giving one decelerating and two accelerating mechanisms. There is a single input $u \in [0, 3]$, so $m = 1 < d - 1 = 2$, and Theorem 4.1 puts zero loss out of reach unless this plant is one of the exceptions of Remark 4.6; Section 6.3 computes the floor and finds that it is not. Noise is unit and the prior variance is $\text{Var}[R] = 10^3$, so the weight $w = \gamma/(1 + \gamma)$ of Lemma 2.2 is evaluated at the plant's own $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ at each state rather than held fixed; γ varies along a trajectory because $\|\mathbf{d}\|$ does. Every trajectory starts from $x_i(0) = 5 \times 10^{-3}$ and runs to the condemnation limit $x_f = 0.9$ of (3). Raising x_f to unity leaves every qualitative conclusion below intact and moves the constants: the peak floor of Section 6.3 rises from 4.52° to 5.41° , and the factor of Section 6.5 at the steering indicator falls from 68 to 58. No constant reported below should be read across thresholds.

Verification of the identities The analytical identities were checked numerically before use, and the residuals are reported in Supplementary Section S4.

6.2 A Fisher level set of varying loss

Proposition 4.7 is easy to demonstrate on this plant. For a constant input, a common gain on $\mathbf{d}$ moves $\|\mathbf{d}\|$ without rotating $\hat{\mathbf{d}}(\tau)$, so the record's score can be equalised across inputs while the misalignment is left to the input. Write $\delta(u)$ for the gain that equalises the score, fixed by $\delta(u)^2 \|\mathbf{d}\|^2(u) = F$ with F its value at $u = 1.5$. It is equalised by rescaling how long the component runs under load, which is why it does not rotate $\hat{\mathbf{d}}$. By construction $\delta = 1$ at $u = 1.5$, and $\delta > 1$ wherever the raw score is smaller, so it is a multiplier on exposure rather than a fraction of it. Table 2 does exactly that, holding the record's score equal to seven digits by tuning the exposure factor δ . The loss still spans a factor 9.4. The indicator's own information, averaged the same way, orders the five inputs exactly as the loss does, which is what (25) leads one to expect.

The two objectives also disagree on which input to prefer: the raw score, before δ equalises it, is largest at

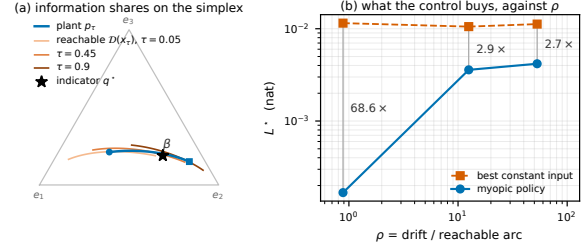


Figure 4: (a) The information-share curve p_τ on the 2-simplex (circle: start, square: failure), the reachable sets $\mathcal{D}(x_\tau)$ at $\tau = 0.05, 0.45$ and 0.9 , and the optimal indicator q^* , which lies off the arcs near birth and failure; zero loss would need it on the arc of every state along $\mathcal{S}_{q^*}$, beginning at birth (Theorem 4.1). (b) What steering buys against the best constant input at the same indicator, plotted against ρ , the drift of the reachable arc over its length.

$u = 0$, which is why δ is smallest there, and the loss is 1.7 times its value at $u = 1$. How far they disagree is a property of the instance; what is structural is only the blindness.

6.3 The reachability floor with a single input

With $m = 1 < d - 1$ no direction lies on every reachable set at once. Along the closed loop at q^* the floor is largest at the start of life, where $\beta_{\min} = 4.52^\circ$; it falls through mid-life to below the 0.01° resolution of our sampling, where the indicator is momentarily reachable, and rises again towards failure (Fig. 5(a)). Zero loss with q^* would need q^* reachable at every point of $\mathcal{S}_{q^*}$, beginning with x_0 , where the floor is at its largest. It is therefore unattainable however the input is steered, by Theorem 4.1(i). Fig. 4(a) shows the geometry: the reachable set sweeps across q^* as the plant ages, so the mid-life arc passes close to the indicator while the arcs at $\tau = 0.05$ and $\tau = 0.9$ stand off it.

6.4 The counting rule, tested in both directions

Theorem 4.1 is an equivalence, and the paragraph above tests only its necessary half: with too few inputs the floor stays positive at x_0 . The sufficient half is the more useful claim for a practitioner deciding how many operating variables to instrument, and it is also the more demanding one, because supplying $d - 1$ inputs is not enough by itself: the aligning input has to stay inside the envelope along the whole of $\mathcal{S}_q$. We test it directly.

Give the same three-mechanism plant a second independent input by writing $\varphi_i(u) = \exp(\theta_i - E_i u_1 - G_i u_2)$ with $G = (0.9, 0.3, 1.2)$, so that $m = 2 = d - 1$ and each input acts on the mechanisms in a different pattern. Then $\log \mathbf{d}_i$ is affine in (u_1, u_2) , and match-

Table 2: Loss of the fixed indicator at a record Fisher score held equal. The exposure factor δ , defined in the text, holds $\delta^2 \|\mathbf{d}\|^2$ fixed and leaves $\hat{\mathbf{d}}(\tau)$ unaffected; $\bar{I}_z = \delta^2 (\mathbf{v}^\top \mathbf{d})^2$ is the indicator’s own information, and rms β the root-mean-square misalignment. Losses use $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ with $\text{Var}[R] = 10^3$.

u	δ	$\delta^2 \ \mathbf{d}\ ^2$	$\bar{I}_z$	rms β (deg)	ℓ (nat)
0.0	0.235	0.4307565	0.4055	11.80	0.0215
1.0	0.628	0.4307565	0.4207	8.98	0.0123
1.5	1.000	0.4307565	0.4170	11.61	0.0208
2.0	1.564	0.4307565	0.4050	15.60	0.0379
3.0	3.769	0.4307565	0.3421	26.98	0.1161

ing a target direction fixes the $d - 1 = 2$ differences of logarithms: two linear equations in two unknowns, with a coefficient matrix of determinant 0.27 here, hence a unique solution at every state.

Take as target the direction the plant occupies at mid-life under the central operating point (1.5, 1.5) of the box. Zero loss with that indicator confines the state to S_q , and at every one of 600 points of that segment the aligning input lies inside the admissible box $[0, 3]^2$, at least 0.52 from its boundary. Along the segment it sweeps u_1 from 0.55 to 2.48 while u_2 stays between 1.32 and 1.70 (Fig. 5(b)). Driving the plant by that feedback keeps the state on S_q to machine precision and the misalignment below 10^{-6} degrees, zero to numerical precision, up to failure. An independent 90×90 grid search over the box along the segment returns a floor of at most 0.152° , which is the resolution of the grid rather than a property of the plant.

The contrast with the single-input case is the content of the theorem: the same mechanisms, the same shapes, the same envelope on each input, and the peak floor moves from 4.52° to zero, with zero loss attained along the whole closed loop, on the addition of one operating variable. On this plant $d - 1$ inputs are enough. In general they are necessary for generic data and not sufficient, and the envelope condition along S_q decides. Within the adopted model this fixes how many independent control degrees of freedom the plant must offer. How those degrees of freedom correspond to physical actuators, and how the mechanisms are individuated in the first place, are modelling questions this paper does not settle. Proposition 4.5 says which plants escape the count, and Remark 4.6 says why a designer should not count on being one of them.

6.5 The value of steering

The policy used throughout this section is the myopic one: at each state it applies the input that minimises β , searched on a grid of 41 points of $\mathcal{U}$. It is a state feedback and ignores the denominator of (26); Section 6.6 measures what that costs. Minimising its loss over the indicator gives q^* , with weight vector $\mathbf{v}^* = (0.466, 0.771, 0.435)$, at which the myopic policy attains 1.68×10^{-4} against 1.15×10^{-2} under the best constant input, a factor of 68.

That comparison holds the indicator at the one

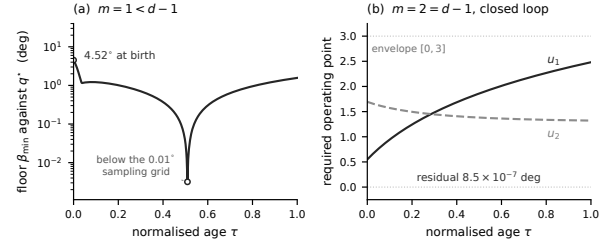


Figure 5: (a) With $m = 1$ the floor against q^* is largest at birth and dips below the sampling grid near mid-life. (b) With $m = 2$ the mid-life direction is held exactly along the closed loop, which runs on the segment S_q ; plotted against the closed loop’s own normalised age is the operating point this requires, which never leaves $\mathcal{U}$.

signed for steering. An operator who will not steer would design an indicator of its own, and minimising over the indicator and the constant input together gives 7.95×10^{-3} , at the hot end $u = 0$ of the envelope. Against that baseline steering is worth a factor of 47, and this is the figure we take as the value of steering on this plant. Both factors depend on how finely $\mathcal{U}$ is discretised for the policy: on 81 points they become 72 and 50, and on 161 points 72 and 50 again.

The factor is a ratio of information losses, not of information, and a large ratio of small losses. By Corollary 2.4, a loss of 7.95×10^{-3} already means an error variance only 1.6% above that of the whole record, and steering removes almost all of that excess. Nor is steering neutral in reliability terms. The myopic loop at q^* lives 1.66 times as long as the best constant input at the same indicator, and 3.8 times as long as the fair baseline, which buys its alignment at the hot end. The myopic loop aimed at a balanced indicator, which calls for far less steering, outlives it in turn: the two differ by a factor 3.4 in lifetime. Aligning a given indicator and prolonging the component are therefore competing objectives, which is the tension anticipated in Section 5 and the reason the horizon cannot be fixed in advance.

6.6 The computed fractional optimum

We solve the discrete Dinkelbach problem at q^* on state grids of 25 to 137 points per axis, with the control grid of the policy. The values converge at first order in the

mesh width: fitted to the four finest grids, the order is 0.99. Extrapolated from those four, $\lambda_{\infty}^* = 1.65 \times 10^{-4}$. The myopic policy attains 1.68×10^{-4} and is realisable, so it bounds the optimum from above, and the optimum lies between the two. The myopic policy therefore sits 1.4% above the optimum as extrapolated. Fitting the four next-finest grids instead, or repeating the computation on control grids of 81 and 161 points, puts the gap between 1.4% and 2.0%, while refining the control grid from 41 to 81 points lowers both the myopic value and the extrapolated optimum by more than 4%.

On this plant, then, the fraction in (26) shapes the statement of the problem and its certification, and not measurably its solution: a policy that attends only to the numerator is optimal to within the resolution of the computation. The lower end of the interval rests on the extrapolation and is not a bound, since even the finest grid returns a value 27% above it. Three checks guard the number: F must vanish at the computed λ^* , λ^* must not exceed a realisable policy, and the grid sequence must extrapolate consistently. A fixed-horizon formulation failed the second and third and was discarded, as did a first attempt with mismatched control grids. A converse bound uniform over trajectories, which would replace this computation by a theorem, remains open.

6.7 Authority, drift, and how far they screen

For a practitioner the first question is not how to steer but whether to bother. The reachable set $\mathcal{D}(x)$ is an arc, and it offers two natural answers: its mean length, which is the actuation authority available to chase the informative direction, and the drift of its midpoint over the life, which is how far that direction wanders. Write ρ for the drift divided by the length, which weighs the two against each other.

Table 3 varies the spread of the activation energies, which by (20) is what sets that authority. As the arc collapses, the attainable loss L^* rises by a factor 25 and the value of control falls from 68 to 2.7 against the best constant input at the same indicator, and from 47 to 1.9 against a constant design with its own: with weak authority there is little left to steer. Fig. 4(b) plots the two against ρ . In this family the drift barely moves, staying between 24° and 30° while the arc spans nearly two decades, so ρ is the reciprocal of the arc in all but name and the table cannot say which of the two matters. Both can be computed from a nominal degradation model before any indicator is designed.

Three plants differing only in their activation energies are not a screening law, so the index was tested on 299 further plants drawn at random: two to five mechanisms, one or two independently adjustable inputs, and the pre-exponentials, activation energies, shape coefficients, input envelope, initial state and signal-to-noise ratio varied together. The gain is measured as in the first gain column of Table 3, against the best constant input at

the steering indicator, which on the three plants of that table overstates it by factors of 1.3 to 1.4 relative to a constant design with its own indicator. The rank correlation between ρ and the gain is -0.36 , with a bootstrap 95% interval from -0.46 to -0.24 , and it keeps its sign inside every stratum of the mechanism count and of the input count, so it is not an artefact of either.

Read as a classifier of whether steering repays a factor of five, which 32% of the plants do, ρ has an area under the receiver-operating curve of 0.73, with interval 0.67 to 0.79. The reachable arc alone scores 0.73 as well, and the drift alone 0.53, with interval 0.46 to 0.60, which includes the 0.5 of a coin toss.

Equal areas do not make the two equally good, and the question is settled by a model that contains both as special cases: a logistic regression on the logarithms of arc and drift as separate terms, of which the arc alone and the ratio ρ are each a constrained case. The drift adds nothing to the arc as a main effect ($p = 0.47$), and constraining the two to enter as a ratio fits significantly worse than the arc alone ($p = 7.7 \times 10^{-5}$; Akaike information 351 against 336). Both results hold in each input count separately. The drift does carry information, but through the arc ($p = 3.0 \times 10^{-4}$ for their interaction), and in opposite directions at the two ends. Among plants of low authority the share repaying fivefold falls from 0.18 to 0.08 as the drift rises from its lowest to its highest third; among plants of high authority it rises from 0.48 to 0.61. Drift that outruns the authority defeats steering, and drift that the authority can follow is what gives steering something to do. The ratio encodes only the first of these. It is also why the lowest tenth of ρ repays fivefold only 43% of the time: there the drift is small, a median of 2.7° against 14° over the sample.

Out of sample the ranking agrees. Over repeated tenfold cross-validation the arc and ρ each rank unseen plants with an area of 0.72, and a model adding the drift and its interaction with the arc reaches 0.74, higher in every repeat. The quantity to screen with is therefore the arc, the authority itself, refined by the drift only through that interaction; ρ remains a convenient description of the family of Table 3, where the drift is nearly fixed. The first row of that table is not typical of its own regime: among sampled plants within a factor 1.4 of its ρ , the median gain is 5.2 rather than 68.

6.8 What the loss buys a downstream estimator

Corollary 2.4 prices the loss in the error of the estimate the model itself prescribes. A practitioner does not use that estimate, so the ordering is worth checking against a procedure that uses none of the construction: sample the indicator over the last 40% of life, match the observed history against the indicator's own reference curve, and read remaining life off the best-fitting time offset. That estimator sees a scalar signal and nothing else — not $\hat{d}$, not γ , not the loss.

Table 3: Authority and drift in a one-parameter family. The activation energies E set the actuation authority; c , the noise and $\mathcal{U}$ are held fixed. L^* is the loss under the myopic policy at the indicator optimised for it. The first gain is its ratio to the best constant input at that indicator, the second to the best constant input with the indicator optimised for it.

E	arc	ρ	L^*	gain	own-indicator gain
(0.60, 1.10, 0.80)	33.9°	0.88	1.7×10^{-4}	68×	47×
(0.80, 0.82, 0.79)	1.93°	12.5	3.6×10^{-3}	2.9×	2.2×
(0.80, 0.805, 0.798)	0.45°	53.1	4.2×10^{-3}	2.7×	1.9×

Over twelve indicators spanning $\bar{\ell}$ from 0.012 to 0.37, its median absolute error — one of the standard prognostic accuracy measures [32] — rises with the loss, at Spearman correlation 0.89, and the best-aligned indicator beats the worst by a factor 1.5. Twelve indicators is a small sample, so both are given with intervals obtained by resampling the 160 units behind each median: 0.34 to 0.90 for the correlation, and 1.40 to 2.03 for the factor. The ordering is not in doubt — a one-sided permutation test gives $p = 1.5 \times 10^{-4}$ — but its strength is, and the point estimate sits near the top of its own interval. The factor is far smaller than the 31-fold spread in $\bar{\ell}$, and it should be: the loss is a logarithmic information quantity, the error is a time, and an estimator with the whole history recovers much of what any one instant discards.

The same protocol on 120 further plants, drawn at random as in Section 6.7, gives correlations between 0.87 and 0.99, with the plant above among the weaker. That design tilts every indicator toward the same direction, however, so loss and error both grow with one parameter and their agreement is close to structural. Tilting instead by a random amount of up to 60° in a random direction weakens the median correlation to 0.74, with the interval excluding zero on 84% of plants. It also exposes what the experiment cannot show. On these plants the angle between an indicator and the mid-life informative direction ranks the indicators identically to the loss on 82% of plants, with neither consistently better on the rest, because the direction rotates little beside how far the indicators differ. The experiment is therefore evidence that the ordering of the loss carries over to an estimator built from none of this construction; it is not evidence that the time-averaged loss improves on a mid-life snapshot of the misalignment, which would need a plant whose direction rotates by more than its candidate indicators differ.

6.9 Robustness to model error

Everything above assumes the designer knows φ_i and h_i . In practice the activation energies are estimated from accelerated tests and carry error, so the question a robust design must answer is how much of the benefit survives on a wrong model. We answer it by direct computation on the test plant rather than by a perturbation bound.

For each level ε we draw a perturbed energy vector $E(1 + \varepsilon\eta)$, with η standard normal, and build the barycentre indicator of Proposition 3.3 from that wrong

Table 4: Cost of designing the indicator from a misspecified stress model. Entries are the loss on the true plant relative to that of the indicator designed from the true model, over 40 draws per level; ε is the relative standard deviation of the error in the activation energies. The steering policy uses the true plant.

ε	median	90th percentile	worst of 40
0.02	1.03	1.52	2.04
0.05	1.40	2.94	6.44
0.10	3.10	10.8	67.7
0.20	10.0	44.9	78.6
0.35	36.0	94.1	195

plant along the trajectory of a constant input, as a designer following this paper would. We then evaluate it on the true plant under the myopic policy, which brings it as close to the informative direction as the envelope allows at each state, with every loss at the plant's own γ . That policy is computed from the true plant, so the table prices an error in the indicator alone; an error that also reached the controller would cost more. Table 4 reports the ratio of that loss to the loss of the indicator designed from the true model, over 40 draws per level, so a ratio of 1 means the error cost nothing.

An error of 2% in the energies is almost free. At 5% the median cost is a factor 1.40. At 10% the median cost is 3.10 and the worst of forty draws costs 67.7, comparable in size to the whole benefit of steering in Section 6.5; beyond that every column grows monotonically, to a median of 36.0 at $\varepsilon = 0.35$. The design therefore tolerates an error of two percent in the energies almost entirely and one of five percent in the median, but not one of ten percent, so the accuracy of the stress model bounds what steering can deliver in practice.

The structural counterpart is the knife edge of Remark 4.6: the plants that escape the counting rule do so only at exact parameter values, and at one percent of error in their shape coefficients steering is already worse than not steering, a fragility that closing the loop halves but does not remove (Supplementary Section S4). Both results point the same way: the counting rule of Theorem 4.1, necessary for generic data, is the one to design against, and the model it is applied to has to be known to within a few percent.

7 What public data can say about the premises

The construction rests on three modelling choices, and each is empirically testable. They do not carry equal weight, and Fig. 6 separates what depends on which. Two are premises about the plant: that the informative direction rotates along the trajectory, and that the input moves it by acting on the degradation, through $\varphi(u)$. The third is the separability of Lemma 4.4, which is what makes the operator’s authority computable offline, and it is tested together with the pairing of channels with mechanisms in (5). Table 5 summarises what six public run-to-failure fleets currently settle. Supplementary Section S1 gives the protocol behind every row, and its Table S1 lists the parameters of the code that produced it.

How a fleet was generated decides what it can test, and for the two turbofan simulations it decides against the actuation premise. C-MAPSS imposes its flow and efficiency losses as exponential functions of time, from a random initial deterioration, and the operating regime enters only through the thermodynamic response of the sensors [33]. N-CMAPSS imposes a linear normal degradation followed by a steeper abnormal one, and the operating history enters only through the onset of the abnormal phase [34]; within a flight the damage does not respond to the operating point at all. A contrast between operating points inside one engine therefore reads the observation map $H(x, u)$ that (5) excludes, not the stress factor $\varphi(u)$.

Four readings qualify the table. The Severson channels are capacity fade, resistance growth and mean temperature, and mean temperature is a monitoring channel rather than a damage state — it is the operating variable itself in the Arrhenius form of Section 2. That row therefore reads the rotation of a direction formed from three measured series, which is what the premise asserts, and does not need (5)’s pairing of channels with mechanisms. On the four measured fleets the obstacle to testing actuation is the design of the fleet and not the physics: the operating point never changes within a unit, or there are too few units. On the two simulated fleets it is the generator, as set out above, so the premise is untested everywhere. What C-MAPSS does show is that the observed direction depends on the regime through the sensor response alone. A fixed indicator’s loss depends on the observed direction however it moves, that is a channel through which an operating point matters for indicator design, but it is one (5) excludes and it is not evidence for $\varphi(u)$. Both the Severson and the C-MAPSS rows estimate the direction by differentiating each channel along normalised life, which reads a time derivative as the sensitivity of Eq. (6) and so presumes the time-shift population of Section 2. And the C-MAPSS contrast is reported against a negative control — fake regimes cut from a single real one, where no contrast exists by construction — because a process

that returned a positive contrast regardless would produce the same table entry.

The consequence is the scope of the results above, and it is wider than separability alone. The damage the N-CMAPSS simulator imposes is constant within every flight, as the protocol of Supplementary Section S1 checks in the files themselves, so the dependence on the operating point that the two-way layout measures belongs to the sensor map and not to a stress factor. What the test rejects is therefore the pairing (5) for raw turbofan sensors; separability of φ on its own is untested there. The pairing is what delivers the positivity of $\mathbf{d}$, so the geometry of Sections 3 and 4 applies to such sensors only once the observation map has been inverted, as Section 2.3 anticipates; where (5) holds it needs only $\varphi_i > 0$ and $h_i > 0$. What separability buys is Proposition 4.5 and the offline route to $\mathcal{B}$: those describe a model class, and a plant must be shown to belong to it before they are used. The two-way layout above is the test, and it costs one pass over data a reliability programme already holds.

8 Discussion

Four practical readings follow from the results above. The first two hold whatever the plant is, because one is a test and the other an identity. The last two assume the separable form of (2) read through (5), which Section 7 reports as rejected jointly on the only data able to test it, so they apply where that form does and not otherwise.

Test the separability assumption before relying on it. The two-way layout of Section 7 needs only a fleet whose operating point varies within a unit, and costs one pass over data a reliability programme already holds. If the interaction clears the measured noise floor, either the stress model or the pairing of channels with mechanisms fails, the offline route to the operator’s authority is unavailable, and the two model-based readings below do not apply.

Score the indicator’s information, not the record’s. This is the second of the two unconditional readings. By Proposition 4.7 the record’s Fisher information does not fix the loss, so an input chosen to maximise a scalarisation of it can leave the loss far from where another input would put it, as Table 2 shows at equal record information. Since the two informations together fix the loss by (25), observability-aware input design serves a fixed indicator once it scores I_z alongside I_{rec} .

The binding constraint is actuation authority, not indicator sophistication. Theorem 4.1 prices the operating variables a lossless fixed indicator would need: fewer than $d - 1$ fail for almost every indicator, and $d - 1$ suffice only where the aligning input stays inside the envelope along the segment the indicator fixes, which Proposition 4.5 turns into a containment on the separable subclass. The exceptions of Remark 4.6 are a knife edge. Care in choosing the weights substitutes for actuation in neither case. What one controllable variable

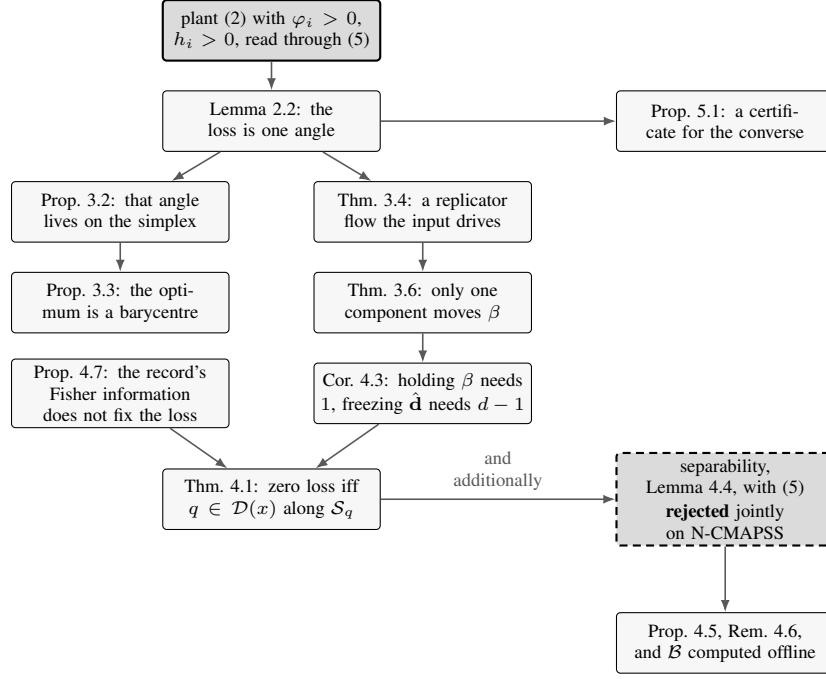


Figure 6: What rests on what. The spine needs only the positivity of $\mathbf{d}$, which the degradation model gives in the channel coordinates of (5). The dashed branch adds separability. Section 7 reports the two rejected jointly on raw turbofan sensors, which keeps those sensors off the spine as well until the observation map is inverted.

does buy is the weaker objective of Corollary 4.3: an installation that could never freeze the informative direction can still hold the misalignment constant, though not at zero. Authority bought this way is not free, and Section 6.5 prices it in life.

The decision to steer precedes the design of the indicator. The width of the reachable arc requires only a nominal degradation model and the operating envelope, and Table 3 shows the value of steering collapsing from $68\times$ to under $3\times$ as it shortens, measured against the best constant input at the steering indicator, and from $47\times$ to under $2\times$ against a constant design with its own. It screens rather than decides, and across random plants the drift of the informative direction refines it only in combination: drift the authority can follow gives steering something to do, and drift that outruns it defeats steering (Section 6.7).

Limitations Four limitations qualify the readings above. The separable form of (2), read through the pairing of (5), is not only an idealisation but one the data reject: Section 7 finds the interaction between operating point and state larger than either main effect on two N-CMAPSS datasets, so computing the operator's authority offline is available only where that form holds. The optimum of Section 5 is bracketed rather than certified, and on the test plant the bracket does not separate it from the myopic policy. The actuation premise is untested on every public fleet, for the reasons set out in Section 7. And the population is taken to slide along one degradation path, which is what makes

remaining life a coordinate on it; under dispersion in the coefficients of (2) the sensitivity that Remark 2.1 measures stands some 45° from (6), so the construction describes fleets that vary in when they fail rather than in how they degrade.

9 Conclusion

A fixed health indicator discards information, and this paper has treated what it discards as a quantity the operating point controls. Indicator design then becomes a reachability question. The family of reachable direction sets bounds what the operating point can achieve, and their width, available before any indicator is fixed, indicates whether substantial benefit is plausible, and across random plants of this model the drift of the informative direction refines that indication only in combination with it. Under the proposed model, actuation authority, rather than sophistication of the indicator, governs whether a fixed compression can come close to lossless. What steering delivers in practice is bounded by how well the stress model is known: an error of two percent in the activation energies is almost free, and one of ten percent is not (Section 6.9). Whether that authority is present is an empirical question rather than a modelling assumption, and no public fleet yet answers it: the simulated turbofans keep their damage independent of the operating point by construction, and the measured fleets do not vary it within a unit. Answering it awaits records in which the operating point varies within units and acts on the damage.

Table 5: What the six public fleets settle, and what they cannot. Only the two turbofan simulations vary the operating point within a unit, and neither lets it act on the damage, so the actuation premise is untested on every public fleet.

fleet	premise	what is measured	reading
Severson lithium-ion [35]	rotation	arc of an empirically constructed informative direction, the fleet mean, over $\tau \in [0.15, 0.9]$, on 129 Severson lithium-ion cells	travels 30.7° of arc and ends 21° from its start (Fig. 2(c)); stays in the open positive orthant once the window reaches 301 cycles
XJTU-SY [36], PRONOSTIA [37] NASA batteries [38]	actuation	—	the operating condition is fixed per unit, so condition and group of units cannot be separated four cells
C-MAPSS [33]	actuation	—	by 7.9° (FD004) and 10.8° (FD002), positive for 436 of 439 engines; fake regimes give $ B - W \leq 0.5^\circ$; the damage is regime-free by construction, so this reads the sensor map
N-CMAPSS DS02, DS01 [39]	actuation: not testable	between-regime minus within-regime angle, each engine serving as its own control	larger by a factor 1.65 over the operating-point effect on DS02 and 1.84 on DS01, under every clustering and staging tried: rejected jointly

Section 7 rejects the separable form, read through the pairing of channels with mechanisms, on the highest fidelity public turbofan data, so the geometry of Section 4.1, and the offline route it opens, describe a model class rather than measured machinery. What replaces that structure when the sensors mix mechanisms or the stress model is not separable, and whether a weaker one still yields a computable reachability criterion, remains open.

Two further questions remain open, the first of them now in a more specific form. Proposition 5.1 reduces the search for a converse bound uniform over trajectories to the exhibition of a single verification function. The value function satisfies it in the viscosity sense, so what is required is a computable version with a certified error, from a grid value function with a residual bound or from a sum-of-squares programming. The other is whether the controlled horizon ever makes the myopic policy measurably suboptimal. On the test plant the computed optimum does not separate from it, and a plant on which it does would show what the fractional structure is worth in practice.

Data availability and reproducibility

The six fleets named in Section 7 are publicly available. One script recomputes every reported number from a single configuration and compares it with the value written in this manuscript and in its supplementary material, covering the tables and the quantities reported alongside them; all of its checks pass, and the output of the run that verified this version is included. The random-plant sweep of Section 6.2 and the grid sequence of Section 6.6 take hours, so they and the downstream repeat of Section 6.8 are recorded by three further scripts; the checks read the records and recompute part of the grid sequence afresh. Three groups of numbers lie outside the checks: the residuals of the identity checks in Supplementary Section S4, the barycentre comparison in Section 3.2,

and the diagnostics of the two candidate certificates in Supplementary Section S3. The plots are produced by three further scripts. All of these are available at <https://github.com/HuyHoangLe0201/health-indicator-information-loss>, and are also provided with this article as a supplementary archive, together with the Supplementary Material.

Supporting information

Additional supporting information is found in the online version of this article.

Supplementary Material (PDF). Supplementary Sections S1 to S4: the protocols behind every row of Table 5 with the parameters of the code that produced them, the proofs of Propositions 2.3 and 3.3, the two candidate certificates for Proposition 5.1 and what each would need, and the numerical verification of the identities and of the exceptional family of Remark 4.6.

Supplementary archive (ZIP). The Supplementary Material and a copy of the repository named in the Data availability section: the script that recomputes and checks every reported number, the records of the two sweeps and the grid sequence it reads, the output of the run that verified this version, and the three figure scripts.

Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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